

0625 | PAPER 4

TOPIC 01

Motion, Forces and Energy

79 topical answers • 2021–2025

Calculations, final values and examiner-keyword summaries

Answers are independently condensed from the official marking requirements. They are not a reproduced Cambridge mark scheme. For drawings and graph lines, compare the stated method with the original question figure.

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Core formulas

- speed = distance $\div$ time
- acceleration = change in velocity $\div$ time
- $F = ma$; $W = mg$
- moment = force $\times$ perpendicular distance
- momentum = mv ; impulse = $F\Delta t$
- pressure = force $\div$ area; liquid pressure = ρgh
- work = force $\times$ distance; KE = $\frac{1}{2}mv^2$; GPE = mgh
- power = energy $\div$ time; efficiency = useful $\div$ total $\times 100\%$

1.1 Scalars, vectors and motion

Q001 • Feb/March 2021 Paper 42 Question 2

1.1 Scalars, vectors and motion

(a)(i) (moment of a force) is the turning effect (about a point / pivot)

(a)(ii) $2.8 \times 10^6 \text{ N m}$

(moment =) Fd in any form

(b)(i) scalar / speed has magnitude only or scalar / speed has no direction

vector / velocity has magnitude and direction

(b)(ii) any scalar quantity

any vector quantity

(c) correct triangle or parallelogram drawn

resultant force (including correct arrow)

scale $1 \text{ cm} = 4 \text{ N}$ or $1 \text{ cm} = 5 \text{ N}$

$40 - 47 \text{ N}$ and $33^\circ - 40^\circ$ (anticlockwise from 20 N)

Q002 • Oct/Nov 2021 Paper 41 Question 1

1.1 Scalars, vectors and motion

(a) it / a vector has a direction

(b) two / three vectors and no more than one other quantity underlined

acceleration and momentum and velocity underlined and no others

(c)(i) 5.5 N

(c)(ii) correct right-angled triangle / rectangle / intersecting arcs seen e.g.

(magnitude from) 9.6 to 10.0 N

(angle to vertical from) 54.0 to 57.5°

(c)(iii) any two of:

equal (in magnitude)

opposite (in direction)

the ring is in equilibrium or no resultant force on ring

or forces on ring balance

Q003 • May/June 2022 Paper 43 Question 3

1.1 Scalars, vectors and motion

(a)(i) suitable scale recorded (e.g. $2 \text{ cm} : 25 \text{ N}$)

two vectors correctly drawn by eye and correct resultant

130 N

(vertically) upwards

(a)(ii) 13 kg

(b) acceleration

momentum

Q004 • Oct/Nov 2024 Paper 43 Question 1

1.1 Scalars, vectors and motion

(a)(i) (scalar) does not have direction or vector has direction

(a)(ii) (momentum =) mass $\times$ velocity

(b)(i) 43 m / s

$F\Delta t = \Delta(mv)$ or $F = \Delta(mv) / \Delta t$ or $F = ma$ or $a = \Delta v / \Delta t$

$(u =) Ft / m$ or $(u =) (27\,000 \times 1.5) / 950$

(b)(ii) straight horizontal line from y axis extending to $t = 4.0 \text{ s}$

straight line from $t = 4.0 \text{ s}$ with negative gradient meeting the x axis (less than half-way between 4.0 and the end of the time

axis)

Q005 • Feb/March 2025 Paper 42 Question 1

1.1 Scalars, vectors and motion

(a) $50\,000 \text{ N / m}$ or $5.0 \times 10^4 \text{ N / m}$

(k =) $F \div x$ or k is (determined from) the gradient (of the line in Fig. 1.1)

(b) Any one valid point(s):

- The graph is a straight line (through the origin)
- The slope / gradient is constant
- The graph / line does not curve

(c) 1300 N

any negative number or indication that force is in opposite direction

(d) (Force is a) vector quantity. Forces have (both magnitude / size and) direction

Q006 • May/June 2025 Paper 41 Question 1

1.1 Scalars, vectors and motion

(a) (only) acceleration and velocity circled

(b)(i) 56 m / s

braille: 55 m / s

(b)(ii) accelerates or speed increases

(then) constant speed

(then) decelerates or speed decreases

constant acceleration or constant deceleration

(b)(iii) 26 000 m or 26 km

braille: 26 125 m

(distance =) area under the speed-time graph or

(distance =) average speed $\times$ time taken

$0.5 \times 80 \times 56$ and 56×400 and $0.5 \times 50 \times 56$

$\times \{400 + 530\} \times 56$

braille: $0.5 \times 100 \times 55$ and 55×400 and $0.5 \times 50 \times 55$

$\times \{400 + 550\} \times 55$

(b)(iv) Any one valid point(s):

- lower acceleration or less acceleration
- lower deceleration or less deceleration
- lower maximum speed or lower average speed or lower constant speed

Q007 • Oct/Nov 2025 Paper 42 Question 1

1.1 Scalars, vectors and motion

(a)(i) scalar has magnitude (size) only or vector has (magnitude and) direction

(a)(ii) mass, energy and temperature

all correct and no additional quantities 2 marks

2 correct and no additional quantities 1 mark

(b)(i) decrease in kinetic energy = increase in gravitational (potential) energy or

$\frac{1}{2} mv^2 = mg(\Delta)h$

$v^2 = 2gh$ or $v^2 = 2 \times 9.8 \times 5.0$ or $v^2 = 98$

$v = \sqrt{98}$ or 9.899

(b)(ii) 0.57 kg m / s (upwards) or 0.57 Ns (upwards)

(momentum =) mv or $5.8 \times 10^{-2} \times 9.9$

1.2 Speed-time graphs and forces

Q008 • May/June 2021 Paper 41 Question 1

1.2 Speed-time graphs and forces

(a)(i) any value from 35 to 43 m / s²

(a =) $(v - u) / t$ in any form or gradient (of line) or $(58 - 50) / 0.20$ or equivalent values from the graph

(a)(ii) 3800 N

(F =) ma in any form or $\Delta p / \Delta t$ in any form

or $76 \times$ candidate's 1(a)(i)

or 760 seen

76 × candidate's 1(a)(i) evaluated or 76 × (candidate's 1(a)(i) + 10) or 76 × (candidate's 1(a)(i)) + 760

(b) (deceleration because) upward force greater than weight or upward resultant force

air resistance decreases (with decreasing speed / with time)

or deceleration decreases or resultant (upward) force decreases

(until / finally) weight equals air resistance; or forces balance

(c) at zero speed there is no air resistance

weight / downwards force is (still) acting; or there is (now) a resultant force (downwards at zero speed)

forces balance at a speed greater than zero

speed cannot decrease / no deceleration once forces balance

Q009 • May/June 2021 Paper 42 Question 1

1.2 Speed-time graphs and forces

(a) same (as density of surrounding air)

(b)(i) falls

(b)(ii) volume decreases

density increases

(c)(i) starts at origin

finishes horizontal by eye

gradient decreasing smoothly to 0

(c)(ii) 10 m / s² (down)

0 any unit

Q010 • Oct/Nov 2021 Paper 42 Question 1

1.2 Speed-time graphs and forces

(a) (acceleration) increases

(b) tangent drawn at 25 s

78 to 82 m / s²

(c) (distance =) area under graph (stated or correct area clearly shown on graph) or (400 × 10) / 2 or (b × h) ÷ 2

2000 m

Q011 • Oct/Nov 2021 Paper 43 Question 1

1.2 Speed-time graphs and forces

(a) 0.0069 m / s²

5 - 7.5

(acceleration =) gradient of graph or $\Delta v / \Delta t$ in any form or

(60 - 42) 60

(b) 48 000 m or 48 km

area under graph

(18 × 7.5 × 60) + (7.5 × 18 × 60) + (15 × 40 × 60)

(c)(i) (force =) 2.0 × 10⁵ N

(F =) ma or 2.3 × 10⁷ × 0.0087 in any form

(c)(ii) there is a backward / drag force or water resistance

Q012 • Feb/March 2022 Paper 42 Question 1

1.2 Speed-time graphs and forces

(a) 2.2 s

(b) Any two valid point(s):

- Line on graph is horizontal / gradient is zero
- (therefore) no acceleration / speed is constant
- (resultant) force causes / is proportional to acceleration

(c) 8.5 ms⁻²

(a =) $\Delta v / t$ in any form or gradient of graph or 12.8 / 1.5 or other suitable values from graph

(1.5, 12.8) both seen or alternative suitable points on the line identified

(d) 0.5 × 12.8 × 1.5 (= 9.56 / 9.6 m) or 6.4 × 1.5 (= 9.6)

(length of ramp) = area under graph (between 0-1.5 s) or average velocity × time

Q013 • May/June 2022 Paper 41 Question 1

1.2 Speed-time graphs and forces

(a) negative acceleration or decrease in velocity

change in velocity per unit time or rate of change of velocity

(b) delay in applying brakes or (human) reaction time or foot not removed from accelerator**(c)(i) gradient or slope****(c)(ii) $20.5 \text{ m/s} \leq \text{answer} \leq 23.5 \text{ m/s}$**

the coordinates at one point on curve (e.g. (0.50, 11))

and (upper) time coordinate $\leq 1.0 \text{ s}$ **(d)(i) air resistance / air friction acts on the car****(d)(ii) air resistance / resultant / resistive force decreases**

and as speed decreases / car decelerates

air resistance / resultant / resistive force decreases / changes

Q014 • May/June 2023 Paper 42 Question 1

1.2 Speed-time graphs and forces

(a)(i) no resultant / net force

no resultant/net moment

(a)(ii) $4.7 \times 10^7 \text{ J}$ or 47 MJ $(\Delta)E_p = mg(\Delta)h$ or $(\Delta E_p =) mg(\Delta)h$ or $(\Delta E_p =) 3200 \times 9.8 \times 1500$ **(b)(i) point, labelled 1, on either of the horizontal sections of the graph (to the left of A or to the left of B)**

point, labelled 2, on the graph between A and the start of the horizontal section of the graph to the left of B

point, labelled 3, on the graph between the start of the curved section to the right of the origin and the start of the horizontal

section of the graph to the left of A

(b)(ii) (initially there is acceleration due to) weight or gravitational force

or unbalanced force / resultant force / downward force

(then) air resistance increases as speed or velocity increases

(as air resistance increases) resultant force downwards decreases or acceleration decreases

constant speed when air resistance = weight / gravitational force

Q015 • Oct/Nov 2023 Paper 42 Question 1

1.2 Speed-time graphs and forces

(a)(i) 4.1 m/s^2 $(a =) (\Delta)v / (\Delta)t$ or $13(.0) / 3.2$ **(a)(ii) (acceleration is) change / increase in velocity per unit time; or rate of change of velocity****(b)(i) straight line joining (0,0) and (3.2,13.0)**

horizontal line from 3.2 s to 12.0 s

(b)(ii) 21 m

area under speed-time graph (between 0 s and 3.2 s)

or average velocity $\times$ time**(c) ($W =) F \times d$** $F = ma$ or $F(\Delta)t = m\Delta v$ $F = (1350 \times 13) \div 2$ or 8775 (N) or $(F =) 1350 \times 6.5$ $W = 8775 \times 13.0 (= 1.1 \times 10^5 \text{ J})$ or $114\,075 \text{ (J)}$ **(d) any sensible suggestion that increases the stopping distance**

explanation (to match suggestion)

Q016 • Feb/March 2024 Paper 42 Question 1

1.2 Speed-time graphs and forces

(a) constant speed

constant / uniform deceleration

stationary

(b)(i) $7.1 \times 10^5 \text{ J}$ or $710\,000 \text{ J}$ or 710 kJ $E_k = \frac{1}{2}mv^2$ or $(E_k =) \frac{1}{2}mv^2$ or $\frac{1}{2} \times 18\,000 \times (8.9)^2$

(b)(ii) 31 000 N or 31 kN

$$W = Fd \text{ or } (F =) W / d \text{ or } 710\,000 / 23$$

$$F = 710\,000 / 23$$

Q017 • May/June 2024 Paper 41 Question 1

1.2 Speed-time graphs and forces

(a) (acceleration is) rate of change in velocity or change in velocity per unit time or (a =) $\Delta v / \Delta t$

(b) 0.021 N

$$F = ma \text{ or } (F =) ma \text{ or } 0.0075 \times 2.8$$

(c)(i) Any four valid point(s):

- (acceleration) decreases
- (acceleration decreases) to zero (at approximately 0.03 s)
- resistive force increases / resistance increases (as speed / velocity increases)
- resultant force (downwards) decreases
- (until) terminal velocity / constant speed (is reached)
- (when) resistive force = weight or resultant force is zero or forces are balanced

(c)(ii) tangent drawn at $t = 0.010$ s

$$.2 \text{ m} / \text{s}^2 \leq \text{acceleration} \leq 1.8 \text{ m} / \text{s}^2$$

(a =) gradient of tangent or (a =) $\{\Delta y / \Delta x\}$

Q018 • Oct/Nov 2024 Paper 41 Question 2

1.2 Speed-time graphs and forces

(a)(i) $1.8 \times 105 \text{ kg m} / \text{s}$ or $1.8 \times 105 \text{ N s}$

$$p = mv \text{ or } (p =) mv \text{ or } 1400 \times 130$$

(a)(ii) (scaled) area under the (graph) line

(a)(iii) 420 m

$$\frac{1}{2}v_{\text{max}}t \text{ or } \frac{1}{2} \times 130 \times 6.5 \text{ or } \frac{1}{2}bh$$

(b)(i) gradient is negative or speed decreases

(b)(ii) gradient is changing or line / graph / it is a curve / curved

(c) (from) kinetic (energy store)

to internal / thermal (energy store as final store)

Q019 • Oct/Nov 2024 Paper 42 Question 1

1.2 Speed-time graphs and forces

(a)(i) $7.3 \times 107 \text{ N}$

(a)(ii) Key physics: weight gravitational force mass / object effect field

mass

(b)(i) (from 0 to A) increasing acceleration

(from A to B) constant / uniform acceleration

(b)(ii) tangent drawn at time = 400 s

$$\Delta y / \Delta x \text{ from candidate's tangent seen and } 17 \text{ m} / \text{s}^2 \leq \text{acceleration} \leq 23 \text{ m} / \text{s}^2$$

(c) resistive force / air resistance / drag increases as velocity increases

Key physics: until gravitational force balanced air resistance terminal velocity resultant / net zero velocity)

Q020 • May/June 2025 Paper 42 Question 1

1.2 Speed-time graphs and forces

(a) constant speed or uniform speed

(b) 33 m

$$(\text{distance} =) \text{area under graph or } bh \text{ or speed} \times \text{time or } 11 \times 3$$

(c)(i) (rate of) decrease of velocity or decrease in velocity (per unit time) or negative acceleration

(c)(ii) tangent drawn at 9.0 s on graph

$$0.50\text{--}0.70 \text{ m} / \text{s}^2$$

correct method of gradient calculation using any two correct pairs of co-ordinates ($\pm \frac{1}{2}$ a small square) from their tangent

Q021 • May/June 2025 Paper 43 Question 1

1.2 Speed-time graphs and forces

(a) 11.5 m / s**(b) 45 m**(distance =) area under graph or $\frac{1}{2}bh$ (+ bh) $\{0.5 \times 20\} + \frac{1}{2}\{20 \times 3.5\}$ or $\{10 + 35\}$ or $\{0.5 + 4\} \div 2 \times 20$ **(c) 5.7 m / s²**(deceleration =) $(\Delta)v \div \Delta t$ or deceleration = gradient $(0 -) 20 \div \{4(0) - 0.5\}$ or $20 \div 3.5$ **Q022 • Oct/Nov 2025 Paper 41 Question 1**

1.2 Speed-time graphs and forces

(a) the velocity is the speed in a particular direction or velocity has a direction or velocity is 56 m / s in a certain direction

velocity is a vector or speed is a scalar

(b) 6.9×10^8 J or 690 000 000 J or 690 MJ(KE =) $\frac{1}{2}mv^2$ or $0.5 \times 440\,000 \times (56)^2$ **(c)(i) 5.3×10^5 N or 530 000 N**(F =) ma or $440\,000 \times 1.2$ **(c)(ii) 1300 m or 1.3 km****(c)(ii) (d =) W / F** or $(\Delta)t = \Delta v / a$ or $56 / 1.2$ or $(\Delta)t = \text{change of momentum} / \text{force}$ (distance =) $6.9 \times 10^8 / 530\,000$ or (distance =) average velocity $\times$ time takenor 1.3×10^3 **1.3 Forces, acceleration and Newton's laws****Q023 • Feb/March 2021 Paper 42 Question 1**

1.3 Forces, acceleration and Newton's laws

(a) 78 N(m=) ρV or $\rho = m / V$ in any form $W = mg$ **(b) 4.5×10^3 N**(F=) $(\Delta)PA$ or $P = F / A$ in any form $(\Delta P = 1.3 \times 10^5 - 1.0 \times 10^5 =) 3 \times 10^4$

outwards

(c) ($\rho =$) 800 kg / m³ $(\rho =) P / gh$ or $P = \rho gh$ in any form $(\rho =) 9.6 \times 10^4 / (10 \times 12)$ **Q024 • Oct/Nov 2021 Paper 41 Question 4**

1.3 Forces, acceleration and Newton's laws

(a)(i) straight line begins at (15 s, 120 m) and continues to end of given line**(a)(ii) curve with increasing gradient from origin to beginning of candidate's (a)(i)****(b) ($E_k =$) $\frac{1}{2}mv^2$ in any form** $\frac{1}{2} \times 1.8 \times 10^5 \times 20^2$ 3.6×10^7 J**(c)(i) (work done =) force $\times$ distance (moved in the direction of the force)**(work done =) force $\times$ distance moved in the direction of the force**(c)(ii) 240 m c.a.o.****(c)(iii) $3.6 \times 10^7 / 240$ or kinetic energy / distance or ($a =$) $20 / 24$ or $\Delta v / t$ in any form or 0.83 or ($F =$) ma in any form** 1.5×10^5 N

Q025 • Oct/Nov 2021 Paper 42 Question 2

1.3 Forces, acceleration and Newton's laws

(a) extension is (directly) proportional to load (if elastic limit is not exceeded)**(b)(i) 0 to 20.5 + / - 0.5 N****(b)(ii) ($k =$) F / x or ($k =$) $1 / \text{gradient}$**

140 N / m or 0.14 N / mm

(b)(iii) 60 or 61 or 62 or 63 (mm) seen

180 mm or 0.18 m

(c) $W = mg$ in any form or ($m =$) W / g or ($m =$) $4 / 8.7$

0.46 kg

Q026 • May/June 2022 Paper 42 Question 2

1.3 Forces, acceleration and Newton's laws

(a) ($F = 2.0 \times 4.0 =$) 8.0 N $(F =) ma$ in any form**(b) ($F = 30 - 12 =$) 18 N** resultant force on 3 kg mass ($3 \times 4 =$) 12 (N)(weight of 3 kg mass = $3 \times 10 =$) 30 (N)**(c)(i) ($\Delta v =$) $4.0 \times 0.80 =$) 3.2 m / s** $(\Delta)v = at$ in any form**(c)(ii) ($t = 0.020 / 3.2 =$) 0.0063 s or $6.3 \times 10^{-3} \text{ s}$** $(t =) d / v$ in any form**Q027 • Oct/Nov 2022 Paper 43 Question 1**

1.3 Forces, acceleration and Newton's laws

(a) $v - u$

change of velocity per unit time or

 t **(b)(i) ($72 / 35 =$) 2.1 m / s^2** **(b)(ii) 230 000 N or 230 kN** $F = ma$ or $(F =) ma$ or $110\,000 \times 2.1$ **(b)(iii) Any one valid point(s):**

- (increase / change in) air resistance
- (increase / change in) wind

(b)(iv) Any three valid point(s):

- initial acceleration highest value and horizontal line
- curved or straight line downwards
- curved or straight line downwards and line not reaching zero by 35 s
- horizontal line before and up to 35 s.

Q028 • May/June 2023 Paper 43 Question 1

1.3 Forces, acceleration and Newton's laws

(a) 1200 N and $g = W / m$ or $(W =) mg$ or $(W =) 120 \times 9.8$ **(b) $F = ma$ or ($a =$) F / m** $(a =) 54 / 120$ **(c) ($v =$) 3.6 m / s** $a = (\Delta)v / t$ or $(\Delta v =) at$ or $(\Delta v =) 0.45 \times 8(.0)$ **(d) ($d =$) 14 m** (3.6) average speed = (total) distance travelled / (total) time taken or 1.8×8 or $() \times 8$ (2) **Q029 • Oct/Nov 2023 Paper 43 Question 2**

1.3 Forces, acceleration and Newton's laws

(a)(i) 3.5 N / cm or 350 N / m

$k = F / x$ or $(k =) F / x$ or $(k =) \text{load} / \text{extension}$ or $14 / 4$

$(k =) 14 / 4$ or $14 / 0.04$ or 3.5×10^4

(a)(ii) mark at end of straight-line portion

(b)(i) arrow below the horizontal and to the left of vertically down from car

arrow radial

(b)(ii) (force) increases

Q030 • Feb/March 2025 Paper 42 Question 4

1.3 Forces, acceleration and Newton's laws

(a)(i) (acceleration is) rate of change in velocity or

(acceleration is) change in velocity per unit time or

(acceleration is) change in velocity per second

(a)(ii) 200 km = 200 000 m or $\times 1000$ seen

division by $\{60 \times 60\}$ seen or division by 3600 seen

(a)(iii) 80 s or 79 s

$(t =) \Delta v / a$ or $56 / 0.7(0)$

(b)(i) 310 000 N or 3.1×10^5 N

$F = ma$ or $440\,000 \times 0.7(0)$

(b)(ii) (statement:) reduces acceleration or lower (maximum) velocity

(explanation:) resultant/net force decreases

Q031 • May/June 2025 Paper 41 Question 2

1.3 Forces, acceleration and Newton's laws

(a)(i) direction (changes)

Any one valid point(s):

- magnitude of the velocity (changes)
- speed (changes)
- (there is) acceleration

(a)(ii) Any one valid point(s):

- (change) size
- (change) shape

(b) 1 place (the centre of) the metre ruler on the pivot

(labelled) diagram showing metre ruler on a pivot

add mass on one side (of pivot) and then add mass on other side to balance the ruler

3 (moment =) force $\times$ perpendicular distance (from pivot)

4 sum of clockwise moments = sum of anticlockwise moments (when in equilibrium)

1.4 Momentum and impulse

Q032 • May/June 2021 Paper 41 Question 2

1.4 Momentum and impulse

(a)(i) 0.078 N s or 0.078 kg m / s

$(I =) m\Delta v$ in any form or 1.2×0.065

(a)(ii) 150 m / s

$v_b = (m_i + v_i) / m_b$ in any form or initial momentum = final momentum

or $1.2(0052) \times 0.065 / 0.00052$ or $0.078(0338) / 0.00052$

(b) work done against / due to / because of friction; or kinetic energy (of trolley) used to do work

kinetic energy decreases (to zero)

thermal energy produced

Q033 • Oct/Nov 2021 Paper 42 Question 3

1.4 Momentum and impulse

(a) momentum before collision = momentum after collision

(initial momentum (p) =) 800×2 or 1600 (kg m / s)

$$(v =) (1600 - 1300) / 800 \text{ or } 300 / 800 \text{ or } 0.38 \text{ (m / s)}$$

(b)(i) (impulse =) change in momentum

$$1300 \text{ N s}$$

(b)(ii) same value as (b)(i) or 1300 (Ns)

Q034 • Oct/Nov 2022 Paper 41 Question 1

1.4 Momentum and impulse

(a) 2.3 J

$$\Delta \text{g.p.e.} = mg\Delta h \text{ in any form or } 0.50 \times 10 \times 0.45$$

(b)(i) 1.2 N s

$$\text{impulse} = \text{change in momentum or } 2.0 \times 0.60$$

$$I = m\Delta v \text{ in any form or } 2.0 \times 0.60$$

(b)(ii)

kinetic energy (of block A) decreases

thermal / internal energy produced / increases (due to friction)

friction mentioned or block slows down / decelerates

Q035 • Oct/Nov 2022 Paper 43 Question 2

1.4 Momentum and impulse

(a)(i) 4.5 kg m / s

$$p = mv \text{ or (change in momentum =) } mv - mu$$

$$(\text{change in momentum} =) (0.058 \times 52) - (-0.058 \times 26) \text{ or } (0.058 \times 52) + (0.058 \times 26) \text{ or } (0.058 \times -26) - (0.058 \times 52)$$

(a)(ii) (impulse =) force $\times$ time or (impulse =) Ft

(a)(iii) 0.013 s

$$(t =) \text{change in momentum} / F \text{ or } (t =) m(v-u) / F \text{ or } (t =) \Delta p / F$$

$$\text{or } 4.5 / 350$$

(b) 59 J

$$KE = \frac{1}{2}mv^2 \text{ or } (KE =) \frac{1}{2}mv^2$$

$$(\text{change in KE}) = \frac{1}{2}0.058 \times 522 - \frac{1}{2}0.058 \times 262 \text{ or } \frac{1}{2}0.058 \times 262 - \frac{1}{2}0.058 \times 522$$

Q036 • Oct/Nov 2023 Paper 41 Question 1

1.4 Momentum and impulse

(a) 20 J

$$(\Delta \text{Ep} =) mg(\Delta)h \text{ or } 0.2(0) \times 9.8 \times 10$$

(b)(i) 14 J

$$(Ek =) \frac{1}{2}mv^2 \text{ or } \frac{1}{2} \times 0.2(0) \times 122$$

(b)(ii) $p = mv$ or $(\Delta p =) mv - mu$

$$(\Delta p =) 0.2(0) \times \{14 + 12\} \text{ or } 0.2(0) \times \{14 - -12\}$$

$$p \text{ before} = 0.2(0) \times 14 \text{ and } p \text{ after} = 0.2(0) \times (-)12$$

$$(\Delta p =) 2.8 - \{-2.4\} (= 5.2 \text{ kg m / s}) \text{ or } (\Delta p =) 0.2(0) \times \{14 - -12\}$$

(b)(iii) 21 N

$$(F =) \Delta p / (\Delta)t \text{ or } F = (\Delta)mv / (\Delta)t \text{ or } 5.2 / 0.25$$

Q037 • Oct/Nov 2023 Paper 42 Question 3

1.4 Momentum and impulse

(a)(i) force $\times$ time (for which force acts)

(a)(ii) 0.056 N s

$$v = s / t \text{ or } v = 0.67 / 0.18 \text{ (m / s)}$$

$$(\text{impulse} =) \Delta\{mv\} \text{ or } (\text{impulse} =) 0.015 \times 0.67 / 0.18 \text{ or}$$

$$(\text{impulse} =) 15 \times 0.67 / 0.18 \text{ or } (\text{impulse} =) 5.6 \times 10 \text{ N}$$

(a)(iii) (momentum is conserved as) air released from the balloon moves in the opposite direction to the balloon

momentum of balloon (and straw) is equal in size to momentum of air

(b) resultant force = 0.84 N resultant force = 0.84 N

correct vector triangle or rectangle drawn use of Pythagoras' theorem

e.g. $a^2 + b^2 = c^2$ or (force $= \sqrt{0.402^2 + 0.742^2}$)

direction 62° (below the horizontal) direction 62° (below the horizontal)

correct resultant force vector with correct arrows on all vectors use of trigonometry to find angle

e.g. $\tan \theta = 0.74 / 0.40$

Q038 • Oct/Nov 2023 Paper 43 Question 3

1.4 Momentum and impulse

(a) 1.3 kg m / s

$p = mv$ or ($p =$) mv or 0.19×6.9 or 190×6.9

or $1.3 \times 10n$

(b) (speed of object $=$) 0.89 m / s or 0.88 m / s

momentum before (collision) = momentum after (collision)

or $1.3 \text{ (kg m / s)} = -(0.19 \times 1.5) \text{ (kg m / s)} + 1.8 v \text{ (kg m / s)}$

or (momentum of object $=$) $1.3 \text{ (kg m / s)} + (0.19 \times 1.5) \text{ (kg m / s)}$

or (momentum of object $=$) $1.3 \text{ (kg m / s)} + 0.29 \text{ (kg m / s)}$

or (momentum of object $=$) 1.6 (kg m / s)

(speed of object $=$) $\{1.3 + (0.19 \times 1.5)\} / 1.8 \text{ (m / s)}$

or (speed of object $=$) $1.6 / 1.8$

(c) (loss of KE $=$) 3.8 J or 3.6 J

$KE = \frac{1}{2} mv^2$ or ($KE =$) $\frac{1}{2} mv^2$ or $\frac{1}{2} \times 1.8 \times 0.892$

((final) KE of object) = 0.70 (J) or 0.71 (J)

or ($\Delta KE =$) $4.5 - (0.2 + \text{calculated KE of object})$

Q039 • Feb/March 2024 Paper 42 Question 2

1.4 Momentum and impulse

(a) (Impulse $=$) force $\times$ time (for which force acts)

(b)(i) $F\Delta t = \Delta\{mv\}$ or ($F =$) $\Delta\{mv\} / \Delta t$ or ($F =$) $\Delta p / t$

$\{2800 \times 1400\} / 1 = 3\,920\,000 \text{ N}$ or $\{2800 \times 1400\} / 1 = 3920 \text{ kN}$

(b)(ii) $4.0 \times 105 \text{ kg}$ or $400\,000 \text{ kg}$

(at maximum mass) force = weight of rocket or $F = mg$

($m =$) F / g or $3.9 \times 106 / 9.8$ or $4(.0) \times 10N \text{ (kg)}$

Q040 • May/June 2024 Paper 43 Question 3

1.4 Momentum and impulse

(a)(i) clockwise moment has increased (and no change to anti-clockwise moment)

(a)(ii) (mass of backpack $=$) $45 / (1.6 \times 9.8)$ 2.9 (kg)

(at balance) sum of clockwise moments = sum of anti-clockwise moments

or $(900 \times 0.85) = 1.6 \times (450 + W)$

(clockwise moment $=$) 765 (N m) or (moment due to backpack $=$) 45 (N m)

or ($W =$) $[\{900 \times 0.85\} - \{450 \times 1.6\}] / 1.6$

(b)(i) (impulse is the) force $\times$ time (for which the force acts) or $I = F \times t$

or (impulse $=$) change in momentum or $\Delta\{mv\}$

(b)(ii) Any three valid point(s):

- change in momentum/impulse is the same (on both floors)
- (change in momentum/Impulse) is over longer time
- force = rate of change of momentum or $F = \Delta\{mv\} / \Delta t$
- less force on child (so less injury)

Q041 • Oct/Nov 2024 Paper 42 Question 2

1.4 Momentum and impulse

(a) (momentum is) mass $\times$ velocity

(b)(i) 1.9 N s or 1.9 kg m / s

impulse = $F\Delta t = \Delta\{mv\}$ or (impulse $= F\Delta t =$) $\Delta\{mv\}$ or (impulse $=$) 0.046×41

(b)(ii) 3800 N

$F = \Delta p / \Delta t$ or ($F =$) $\Delta p / (\Delta t)$ or ($F =$) $\Delta(mv) / (\Delta t)$ or ($F =$) impulse / (Δt) or

$1.9 / 0.0005$ or $3.8 \times 10N \text{ N}$

Q042 • Feb/March 2025 Paper 42 Question 2

1.4 Momentum and impulse

(a) (statement momentum is) zero / 0 and(explanation) momentum is mass $\times$ velocity or the velocity (of B) is zero**(b) 1.1 kg**

Conservation of momentum or momentum after (collision) = momentum before (collision)

or $\{0.75 \times 0.44\} = \{0.75 + m\} \times 0.18$ momentum before collision = 0.75×0.44 (+ $m \times 0 = 0.33$) or momentum after collision = $(0.75 + m) \times 0.18$ **(c)(i) 0.13 N** $(F =) \Delta p \div (\Delta)t$ or $(0.75 + 1.1) \times 0.18 \div 2.6$ or $(F =) 0.33 \div 2.6$ $(F =) \{0.75 + 2(b)\} \times 0.18 \div 2.6$ **(c)(ii) (time taken) Increases**use of $F(\Delta)t = \Delta\{mv\}$ or $F \propto 1 / (\Delta)t$ or same momentum change required**Q043 • Oct/Nov 2025 Paper 43 Question 2**

1.4 Momentum and impulse

(a) (-) 3.5 kgm / s $(\Delta p =) m(\Delta)u$ or $m\{v - u\}$ or $0.14 \times \{25 (- 0)\}$ **(b) 360 N**0.0097 (s) or 9.7×10^{-3} (s) or 3.6×10^{-3} (s) $(F =) (\Delta)p / t$ or $3.5 / 0.0097$

1.5 Circular motion and forces

Q044 • Oct/Nov 2023 Paper 42 Question 2

1.5 Circular motion and forces

(a)(i) evaporation**(a)(ii) air is drier**

because water vapour has condensed / turned back to liquid in the condenser

(b)(i) gravitational (force) or weight**(b)(ii) (force is) perpendicular to the motion (of the clothes)****(c) uses (solar / wind) energy which is renewable; or energy (re)sources not used to generate electricity or**

greenhouse gases not produced or does not use (fossil) fuels

Q045 • Oct/Nov 2025 Paper 42 Question 2

1.5 Circular motion and forces

(a)(i) 7400 N $a = \Delta v / \Delta t$ or $a = \{22 - 13\} / 4.5$ or 2.0 (m / s²) SEEN $(F =) ma$ or 3700×2.0 **(a)(ii) air resistance or drag**

friction (between tyres and road)

(b)(i) arrow from car pointing towards the centre of the circle**(b)(ii) Any two valid point(s):**

- car has higher speed
- less friction (between car and track) or not enough friction
- larger mass

1.6 Density

Q046 • Feb/March 2023 Paper 42 Question 2

1.6 Density

(a) ship is not solid steel / there are air spaces in ship

(average) density of ship is less than the density of the water

(b) the centre of gravity is lower and (so) the ship is more stable

the centre of gravity is lower or ship more stable

(c) $1.4 \times 10^7 \text{ J}$ or 14 MJ or $14\,000 \text{ kJ}$

$\Delta E_p = mg(\Delta)h$ or $(\Delta E_p =) mg(\Delta)h$ or $30\,000 \times 9.8 \times 48$

Q047 • Oct/Nov 2023 Paper 43 Question 1

1.6 Density

(a) (oil) stays on surface / floats on the water

(oil has) lower density than water

liquids of lower density float on liquids of higher density

(b)(i) measure (initial) volume of liquid / water and immerse object

immerse object in a known / measured volume of liquid / water

subtract initial / start volume from final / new volume

or calculate the difference in volume

or measure change in volume

or (alternative answer)

fill displacement can / container with water and immerse object

measure volume of displaced water

(b)(ii) 220 g

$\rho = m / V$ or $(m =) \rho V$ or 2.7×83

$(m =) 2.7 \times 83$ or 2.2×10^3

Q048 • Oct/Nov 2024 Paper 41 Question 3

1.6 Density

(a) force $\times$ perpendicular distance (from pivot)

Any one valid point(s):

- force $\times$ distance from pivot
- reference to perpendicular distance (from pivot)
- reference to perpendicular force

(it measures the) turning effect (of a force)

(b)(i) point where (all) the weight (of an object) seems to act

(b)(ii) $2.2 \times 10^7 \text{ N m}$

1.7 (m) or 3.4 / 2 seen

(c) resultant force = 0 or (all) forces cancel out

resultant moment = 0 or moments balance

Q049 • May/June 2025 Paper 42 Question 2

1.6 Density

(a) (moment =) force $\times$ perpendicular distance (from the pivot)

(b) X and 50 cm labelled to the right of the pivot and left of 80 cm

(c)(i) sum of clockwise moments = sum of anticlockwise moments

$\{mg \times 8\} + \{0.12 \times 38\} = \{0.34 \times 32\}$

Any one valid point(s):

- $mg = [\{0.34 \times 32\} - \{0.12 \times 38\}] \div 8$
- $mg = \{10.88 - 4.56\} \div 8$
- $mg = 6.32 \div 8$
- $mg = 0.79$
- (mass =) $0.79 \div 9.8$

(c)(ii) 520 kg / m^3

(density =) mass $\div$ volume or $0.081 \div \{1(.0) \times 2.6 \times 10^{-2} \times 6(.0) \times 10^{-3}\}$

1.7 Pressure

Q050 • May/June 2021 Paper 41 Question 3

1.7 Pressure

(a) molecules (already very) close / touching

(repulsive) forces (very) large

(b)(i) $6.5 \times 10^5 \text{ Pa}$

(p =) F / A in any form or $8800 / 0.016$ or (Fair =) $1.0 \times 10^5 \times 0.016$

5.5×10^5 or $5.5 \times 10^5 (+ 1.0 \times 10^5)$ or $(1600 + 8800) / 0.016$

(b)(ii) pressure due to (increased height of) oil in cylinder mentioned; or pressure (in liquid) increases as depth increases

to keep the upwards force constant or to lift the (extra) oil

or to counteract / oppose the increased pressure / force / weight of the oil

(b)(iii) (initial) force has to be greater than 8800 N to start the motion

or the upwards force (just) balances the weight (so no movement); or piston / oil has weight parts)

Q051 • May/June 2021 Paper 43 Question 2

1.7 Pressure

(a)(i) pressure = force/area

P inversely proportional to area

same force exerted by each group of books

Key physics: area contact bookshelf group B greater smaller

(a)(ii) (pressure =) 1900 Pa

force = $6 \times 0.52 \times 10$ or $31(.2)$ seen

area = $6 \times 0.013 \times 0.21$ or $0.016(38)$ seen or $163.8 \text{ (cm}^2\text{)}$

(b) (depth =) 19 m

$p = \rho gh$ or $(3.0 - 1.0) \times 10^5 = 1030 \times 10 \times h$ in any form

$h = (3.0 - 1.0) \times 10^5 / 1030 \times 10$ or $h = 2.0 \times 10^5 / 1030 \times 10$

Q052 • May/June 2022 Paper 41 Question 2

1.7 Pressure

(a) gravitational potential energy**(b)(i) $1.6 \times 10^6 \text{ Pa}$**

(p =) $hp g$ (in any form) or $150 \times 1000 \times 10$ or 1.5×10^6

1.5×10^6 or $1.0 \times 10^5 + \{150 \times 1000 \times 10\}$ or $1.0 \times 10^5 + 1.5 \times 10^6$

or $1.6 \times 10^6 \text{ N}$

(b)(ii) $5.6 \times 10^6 \text{ N}$

(F =) pA (in any form) or $1.6 \times 10^6 \times 3.5$

(c) speed (of water) remains constant

otherwise density would decrease or gaps would appear in the water

or volume / density does not change; or liquids incompressible

remains constant

Q053 • Oct/Nov 2022 Paper 42 Question 1

1.7 Pressure

(a) (depth =) 0.211 m

$\rho = m / V$ or $(V =) m / \rho$ or $800 / 1020$

$V = l \times w \times d$ or $(d =) V / (l \times w)$ or $V \div 3.72$

(b) ($\Delta GPE =$) $56(.0) \text{ J}$

$GPE = mg\Delta h$ or $(GPE =) mg\Delta h$ or $(800 / 60) \times 10 \times 0.42(0)$

(mass per second =) $800 / 60 \text{ (kg)}$ or their GPE per minute $\div 60$

(c) ($P =$) 8200 Pa

(P =) hpg

(P =) $1020 \times 10 \times 0.8(00) \text{ (Pa)}$

(P =) F / A

$F = mg$ or

$F = 1020 \times 0.8(00) \times 3.72 \times 10$

Q054 • May/June 2023 Paper 42 Question 3

1.7 Pressure

(a) (force of gravity / weight of person is spread over a much) greater area

$$p = F/A \text{ or } p \propto 1/A$$

(force is same so) pressure is lower (so ice is less likely to crack)

(b) $5.8 \times 10^3 \text{ Pa}$

$$p \text{ (due to water)} = \rho gh \text{ or } (p =) \rho gh \text{ or } (p =) 1000 \times 9.8 \times 0.45 \text{ or } (p =) 4410$$

$$W = mg \text{ or } (W =) mg \text{ or } (W =) (690 \times 9.8) \text{ or } (W =) 6762 \text{ or } (p \text{ (due to ice)} =) 1352.4$$

(pressure =) candidate's calculated pressure due to water + candidate's calculated pressure due to ice

$$\text{or total pressure} = [1000 \times 9.8 \times 0.45] + [(690 \times 9.8) / 5.0]$$

$$\text{or total pressure} = 4410 + 1352.4$$

Q055 • May/June 2023 Paper 43 Question 2

1.7 Pressure

(a)(i) (pressure is) force per unit area**(a)(ii) (variation with depth) increases (as depth increases)**

(variation with density) increases (as density increases)

(b) Any two valid point(s):

- geothermal
- nuclear
- tidal

(c)(i) (statement) non-renewable / not renewable / no

(explanation) (nuclear) fuel is used up

(c)(ii) (statement) renewable / yes

(explanation) waves will always continue; or produced by wind which will always continue

1.8 Moments and equilibrium

Q056 • May/June 2021 Paper 42 Question 2

1.8 Moments and equilibrium

(a) force $\times$ perpendicular distance from pivot / point**(b) ($F_1d_1 = F_2d_2 \Rightarrow 500 \times 20 = F \times 12$)**

numbers substituted in any form

$$(F = 10\,000 / 12 \Rightarrow) 830 \text{ N}$$

(c) clear diagram or description (of object) with pivot

and vertical forces / weights / masses / cord tension

causing moments in each direction

indicate / measure forces and perpendicular

distances

calculates a moment or shows / describes how to

and confirms equality of total moment (in each direction)

and statement of equilibrium / balance

Q057 • Oct/Nov 2022 Paper 41 Question 3

1.8 Moments and equilibrium

(a) 620 N**(b)**

no resultant force (on object in equilibrium)

no resultant moment (on object in equilibrium)

(c)(i) 560 N m

$$(M =) F \times \text{perpendicular } r \text{ or } 620 \times 0.90$$

(c)(ii) 540 N

use of any moment

$$T \times 1.2 \sin 60^\circ (= 560) \text{ or } (T =) 560 / (1.2 \times \sin 60^\circ)$$

1.9 Springs and Hooke's law

Q058 • May/June 2021 Paper 43 Question 1

1.9 Springs and Hooke's law

(a) (extension =) 15 cm

$F = kx$ or $x = F/k$ or $3.0/0.2$

(b) extension is proportional to load

up to the limit of proportionality, extension proportional to load

(c) graph initially straight line with positive gradient that passes through the origin

point labelled, increasing gradient to the right

(d) • from elastic / strain energy

• to gravitational potential energy

EITHER:

• to kinetic energy, when moving from A to equilibrium

or from kinetic energy, when moving from equilibrium to B

Q059 • Feb/March 2022 Paper 42 Question 2

1.9 Springs and Hooke's law

(a)(i) Key physics: extension spring directly proportional force / load applied up limit proportionality

(a)(ii) $W=mg$ in any form or force is (directly) proportional to mass

(a)(iii) 80 N

(b) straight line through / from origin with positive gradient up to 175 N

smooth curve after 175 N with increasing positive gradient

(c) $(80 \text{ N} \times 3.5 \text{ m}) = 280 \text{ J}$

$\Delta E = Fx$ in any form or $GPE = mgh$ in any form

Q060 • May/June 2024 Paper 41 Question 2

1.9 Springs and Hooke's law

(a)(i) 0.16 m / s

conservation of momentum or $m_P v_P = m_Q v_Q$ or $2.7 \times v_P = 1.2 \times 0.36$ or $(m_Q v_Q =) 0.432$ seen

$(v_P =) m_Q v_Q / m_P$ or $1.2 \times 0.36 / 2.7$

(a)(ii) 0.078 J

$(k.e. =) \frac{1}{2}mv^2$ or $(k.e. =) \frac{1}{2} \times 1.2 \times 0.362$

$(k.e. =) \frac{1}{2} \times 1.2 \times 0.362$

(b) (from) elastic (energy store in the compressed spring)

to kinetic (as final energy store of trolleys)

Q061 • Oct/Nov 2024 Paper 43 Question 2

1.9 Springs and Hooke's law

(a) hang mass / weight on the bottom of the spring

use a ruler / metre rule and measure / calculate extension

use of $k = F / x$ to calculate the spring constant

or k is the gradient of load-extension graph

any one of:

- repeat measurements using different masses / weights and plot a load against extension graph
- measure final length and initial length of spring and subtract to obtain (magnitude of) extension / change in length
- use a pointer on the bottom of spring to obtain more accurate measurements on ruler

(b)(i) maximum load that can be applied when the extension is proportional to load

the maximum force up to which the extension is proportional to the load

(b)(ii) (extension of spring $Q =$) 15 cm or 0.15 m

and (extension of spring $R =$) 110 cm or 1.1 m

(b)(iii) Q and extension produced by Q is the right length to baby to just reach the floor with feet

Key physics: Q extension produced R too big gentle bounces space provided would make baby collapse

on floor (so unsafe); or bouncer would have to be hung higher if R was used

1.10 Work, energy, power and efficiency

Q062 • Oct/Nov 2021 Paper 42 Question 4

1.10 Work, energy, power and efficiency

(a) (statement) renewable

(explanation) (wind) is) replaced / replenished; or does not run out

(b) Any two valid point(s):

geothermal

nuclear

tidal

(c) chemical

gravitational potential

Q063 • May/June 2022 Paper 42 Question 3

1.10 Work, energy, power and efficiency

(a) scale at least 2 cm : 1 m / s stated

2.5 m / s and 4.0 m / s vectors correctly drawn by eye and correct resultant

magnitude of resultant velocity = 2.3 - 2.8 m / s inclusive

direction 35° - 40° inclusive (downstream)

(b) ($E = \frac{1}{2} \times 65 \times 2.52 =$) 200 J

($E =$) $\frac{1}{2} mv^2$ in any form

Q064 • Oct/Nov 2022 Paper 42 Question 3

1.10 Work, energy, power and efficiency

(a)(i) Any one valid point(s):

- fossil fuel / named fossil fuel
- biofuel / wood / crops
- hydro
- wave
- wind
- solar cell / panel.

(a)(ii) geothermal or nuclear

(b)(i) yes or it is renewable

tides are continuous / regular / happen every day / always there /; or Moon / Sun always there

consumed / used up or tides are an unlimited resource

(b)(ii) (power =) 4800 W

$KE = \frac{1}{2}mv^2$

($P =$) E / t or ($P =$) KE / s or ($KE / s =$) $\frac{1}{2} \times 6(.0) \times 10^3 \times 2(.0)^2$

electrical (output) power = 40% of KE / s or $0.4 \times 12\,000$

Q065 • Feb/March 2023 Paper 42 Question 3

1.10 Work, energy, power and efficiency

(a) energy cannot be created or destroyed

energy can be transferred / transformed (between energy stores)

(b)(i) energy transferred in one hour at a rate of transfer of 1 kW

(b)(ii) 7200 (kWh)

(Δ) $E = Pt$ or (ΔE) = Pt or 1800×4.0 or 1.8×4.0 or 7.2×10^n

(c) Any two valid point(s):

- geothermal
- nuclear
- tidal

Q066 • May/June 2023 Paper 42 Question 2

1.10 Work, energy, power and efficiency

(a) 26 J

$EK = \frac{1}{2}mv^2$ or ($EK =$) $\frac{1}{2}mv^2$ or ($EK =$) $\frac{1}{2} \times 0.16 \times (18)^2$

$$(E_k =) \frac{1}{2} \times 0.16 \times (18)^2 \text{ or } (E_k =) \frac{1}{2} \times 0.16 \times 324 \text{ or } (E_k =) 2.6 \times 10^3$$

(b) 24 N

$$Ft = \Delta mv \text{ or } F = ma \text{ or } (F =) (0.16 \times 18) / 0.12$$

(c) longer time (of impact / contact) and smaller force (on them)

or longer time (of impact / contact) and does not hurt as much

Q067 • Oct/Nov 2023 Paper 41 Question 5

1.10 Work, energy, power and efficiency

(a) Any three valid point(s):

- description of how the (energy from) water is released
- mention of transfers between energy stores
- (moving) water turns turbine
- turbine turns / drives generator
- name of method to match description

(b) advantage of generating electricity from energy stored in water

disadvantage of generating electricity from energy stored in water

(c) Any two valid point(s):

- geothermal (energy / power)
- tidal (energy / power)
- nuclear (energy / power)

Q068 • Oct/Nov 2024 Paper 42 Question 3

1.10 Work, energy, power and efficiency

(a) Any two valid point(s):

- fossil fuels / named fossil fuel
- biofuels / named biofuel / biomass
- wave(s)
- wind (turbines)
- Hydroelectric
- solar panels / solar cells

(b)(i) $E_k = \frac{1}{2} mv^2$

$$(m =) 11000 \times 2 \div [6.3] \text{ or } (m =) 11000 \times 2 \div [39.69] \text{ or } 554 \text{ (kg) SEEN}$$

(b)(ii) 55 (%)

$$\text{efficiency} = \text{useful energy out(put)} / (\text{total} \text{ energy in(put)}) \times 100 \%$$

$$(6000 \div 11000) \times 100$$

(c) (tides) follow a regular / predictable pattern or (tides) are reliable for wind

Q069 • May/June 2025 Paper 41 Question 3

1.10 Work, energy, power and efficiency

(a) gravitational potential to kinetic to gravitational potential

to thermal (store) or to internal (store)

(b)(i) 1800 J

$$(W =) Fd \text{ or } (W =) 65 \times 9.8 \times 2.8$$

(b)(ii) $F = \Delta p \div (\Delta)t$ and Δp is change in momentum, $(\Delta)t$ is time (taken)

$$F = \Delta\{mv\} \div (\Delta)t \text{ and } \Delta\{mv\} \text{ is change in momentum, } (\Delta)t \text{ is time (taken)}$$

force = rate of change in momentum

force = change in momentum divided by time (taken)

$$F = \Delta p \div (\Delta)t \text{ or } F = \Delta\{mv\} \div (\Delta)t \text{ or } F = I \div (\Delta)t$$

Q070 • May/June 2025 Paper 42 Question 4

1.10 Work, energy, power and efficiency

(a)(i) container A or container with (dull) black (surface)

(dull) black (surfaces) are better emitters of infrared / radiation (than shiny white surfaces)

container A or container with (dull) black (surface)

(dull) black (surfaces) are better emitters (than shiny white surfaces)

(a)(ii) rate of transfer of energy from the container is equal to rate of transfer of energy to the container

(a)(iii) convection**(b) Any three valid point(s):**

- metal is a good / better (thermal) conductor; or wood is a poor conductor
- metal contains free / delocalised electrons or wood does not contain free / delocalised electrons
- electrons carry (thermal) energy through metal or electrons collide with (distant) ions
- wood only transfers (thermal) energy by lattice vibrations

Q071 • Oct/Nov 2025 Paper 41 Question 3

1.10 Work, energy, power and efficiency

(a) (electromagnetic) radiation / light (from the Sun)

(produces) electrical (work done)

(b) (black) is a good absorber / poor reflector (of radiation)**(c)(i) 0.54 kW or 540 W**(output power =) total power input $\times$ efficiency ($\div 100$)or (output power =) $0.72 \times 75 \div 100$ or 5.4×10^3 **(c)(ii) the amount of (electrical) energy transferred by a 1 kW appliance in 1 hour**

or energy transferred in one hour at a rate of transfer of 1 kW

(c)(iii) 7

Any one valid point(s):

- $E = Pt$
- energy produced by one cell per year or $3(c)(i) \times 6 \times 365$
- total power output required or $7400 \div \{365 \times 6\}$
- household energy used per day or $7400 \div 365$
- energy produced by one cell per day or $3(c)(i) \times 6$

(c)(iii) Any one valid point(s):

- household energy used per year $\div$ energy produced by one cell per year
- total power output required $\div$ power output of one cell
- household energy used per day $\div$ energy produced by one cell per day

1.11 Motion, forces and energy - mixed**Q072 • May/June 2021 Paper 42 Question 3**

1.11 Motion, forces and energy - mixed

(a) (PE loss =) mgh and (KE gain =) $\frac{1}{2}mv^2$

PE (loss) = KE (gain)

alternative route 1 for 1st two m.p.s

$$v^2 = u^2 + 2as$$

$$u = 0$$

alternative route 2 for 1st two m.p.s

$$s = ut + 0.5at^2 \text{ or } h = 0.5gt^2$$

$$u = 0 \text{ and } t = \sqrt{3} \text{ or } 1.73$$

$$v^2 (= 2gh) = 2 \times 10 \times 15 \text{ or } v^2 = 300 \text{ or } v = 10\sqrt{3} \text{ or } v = 10 \times 1.73$$

$$\{v = 17 \text{ m/s and } v^2 = 300 \text{ or } v = 10\sqrt{3}\} \text{ or } v = 17.3(2) \text{ m/s}$$

(b) (F =) change of p / (change of) time or rate of change of momentum

$$(F =) 30 \times 17.32$$

$$(F =) 520 \text{ N}$$

Q073 • May/June 2021 Paper 43 Question 3

1.11 Motion, forces and energy - mixed

(a) thinking time is constant**(b) kinetic energy**

$$\text{kinetic energy} = \frac{1}{2}mv^2$$

$$\text{work done (to lose KE)} = Fd$$

(so stopping distance is proportional to v^2)

or (alternative route)

time to decelerate is proportional to v

$d = \text{average } v \times t = \frac{1}{2} v \times t$

d is proportional to v^2

(c)(i) 0.68 s

$t = d/v$ or $15/22$ in any form

(c)(ii) 15 000 N

$Ft = \text{change in momentum}$ or $F \times 2.1 = 1400 \times 22$ in any form

or $F = ma$ or $(F =)(1400 \times 22)/2.1$

Q074 • May/June 2024 Paper 42 Question 1

1.11 Motion, forces and energy - mixed

(a) 0.077 kg or 77 g

$g = W / m$ or $(m =) W / g$ or $0.75 / 9.8$

(b)

use of Pythagoras' theorem e.g. $a^2 + b^2 = c^2$

2 vectors at right angles or

or (force $=$) $\sqrt{(1.22^2 + 0.752)}$

.4 (N)

58(°)

resultant force including correct direction of arrow or use of trigonometry to find angle e.g. $\tan \theta = 1.2 / 0.75$

(c) Any two valid point(s):

- velocity
- speed
- direction
- acceleration / deceleration
- moment

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1.11 Motion, forces and energy - mixed

(a)(i) constant acceleration

(a)(ii) area under the graph or $\frac{1}{2}bh$ or average speed $\times$ time

$0.5 \times 5.9 \times 0.6$ or 1.77 (to ≥ 3 sf)

(a)(iii) 9.5 J

$(\Delta E_p =) m g (\Delta)h$ or $0.54 \times 9.8 \times 1.8$

(a)(iv) candidate's 1(a)(iii) value $\times$ 0.5 (2sf)

(b) Any three valid point(s):

- (initially there is acceleration due to) an unbalanced force; or resultant force gravitational force acting on the ball
- as speed increases the air resistance increases
- (as air resistance increases) resultant force downwards decreases or acceleration decreases
- (eventually) ball falls with constant speed when air resistance = weight or ball falls with terminal velocity when air resistance = weight

(c) (some of the) energy is:

transferred to the thermal energy (store) of the ground (when ball hits ground)

or converted / transferred to thermal energy (store) of the air

or used to deform ball / transferred to the elastic energy (store) of the ball (as it contacts the ground)

gravitational potential energy (in the store at C) will be less (than at A)

(ball has) less kinetic energy (in its store just after B)

1.12 Forces and weight

Q076 • Feb/March 2024 Paper 42 Question 3

1.12 Forces and weight

(a)(i) $2.0 \times 105 \text{ Pa}$ or $200\,000 \text{ Pa}$ or 200 kPa $(P =) F / A$ or $13\,000 / (0.016 \times 4)$ **(a)(ii) Any four valid point(s):**

- friction between road and tyre
- temperature of air / tyre increases
- particles (of air in tyre) move faster
- Key physics: particles air collide harder walls tyre more frequently the tyre)
- force (on the tyre wall) increases, and area (of tyre) is constant (so tyre pressure increases)

(b) (maximum number =) 34 $pV = \text{constant}$ or $\{2.0 \times 106 \times 0.026\} = 1.0 \times 105 \times V_2$ $V_2 = 0.52 \text{ (m}^3\text{)} \text{ or } (V_2 =) \{2.0 \times 106 \times 0.026\} / 1.0 \times 105$ or (number of balloons =) $\{2.0 \times 106 \times 0.026\} / \{1.0 \times 105 \times 0.015\}$ **1.13 Momentum and forces****Q077 • May/June 2025 Paper 43 Question 2**

1.13 Momentum and forces

(a) ($p =$) mv or mass $\times$ velocity**(b) 7790 m / s**

momentum before (collision) = momentum after (collision)

or $m_c v_c + m_{nv} v_n = 6.55 \times 106$ or (momentum of cone =) 120×7850 or 9.42×105 $720v + \{120 \times 7850\} = 6.55 \times 106$ or (momentum after collision =) $6.55 \times 106 - 9.42 \times 105$ or 5.6×106 **(c) $12\,000 \text{ kg}$** $120\,000 \text{ (N)} \text{ or } (m =) W \div g \text{ or } (m =) 1.2 \times 10^5 \div 9.8$ **1.14 Motion and velocity****Q078 • May/June 2023 Paper 41 Question 1**

1.14 Motion and velocity

(a)(i) (magnitude of velocity =) 0.90 m / s

(

use of Pythagoras' theorem e.g. $a^2 + b^2 = c^2$ or (speed =) $\sqrt{0.542^2 + 0.722^2}$

correct vector triangle or rectangle drawn

(direction of velocity =) 53° (to riverbank)use of trigonometry to find angle e.g. $\tan \theta = 0.72 / 0.54$

(only) angle with horizontal identified on the diagram

(a)(ii) (distance =) 81 m $v = s / t$ or ($s =$) vt or ($s =$) $0.9(0) \times 90$ (time =) $1.5 \times 60 (= 90)$ or (time =) 90 **(b) friction (of water backwards) or resistance (on swimmer backwards)**

(friction / resistance) balances forward force or (there is) no resultant force

1.15 Projectile motion and forces**Q079 • May/June 2025 Paper 43 Question 3**

1.15 Projectile motion and forces

(a) ($F =$) $W \div d$ or $W = Fd$

$110 \div 0.45$ or 244 (N)

(b)(i) 86 m / s

$E_k = \frac{1}{2} mv^2$ or $110 = \frac{1}{2} mv^2$ or $110 = \frac{1}{2} \times 0.03 \times v^2$

$(v =) \sqrt{[2 \times 110] \div 0.03}$ or $v^2 = [2 \times 110] \div 0.03$ or $v^2 = 7333$

(b)(ii) air resistance / drag / friction

energy lost to surroundings / energy transferred to thermal / internal energy